# TEST 3

## READING PASSAGE 1

*You should spend about 20 minutes on **Questions 1-13,** which are based on Reading Passage 1 below.*

# Sleeping on the Job

*Can curling up under your desk, or in a purpose-built sleep pod, for a 10-minute sleep improve your performance at work?*

There are times, typically in the afternoon, when many office workers experience a feeling of tiredness and may even drift off to sleep in front of their computers. Many workplaces consider artificial stimulation, provided by coffee or a chocolate bar, more acceptable than a short sleep when attempting to combat this daytime sleepiness. However, there is considerable evidence that trying to work during a spell of daytime drowsiness can be costly. "Workplace accidents and errors peak at the same time that our circadian rhythms (sleep-wake cycle) cause a drop in alertness," says Dr. Gerard Kennedy, a sleep specialist based in Melbourne, Australia. "That's between about two and five pm," he says. These biologically based downturns in alertness are natural and occur even if you've had a good night's sleep. Most workers simply continue during the after-lunch decline or reach for the nearest energiser: a strong coffee, a can of high caffeine soft drink, a cigarette or some secretly stored chocolate in the top drawer.

However, a growing number of workers are taking very short sleeps, or "power naps" instead. "Research shows that a nap can improve your mood and productivity, alleviate tiredness, increase alertness and reduce errors at work," says Kennedy. "A nap as brief as 10 minutes will produce these results." It seems that the length of the nap is significant. Professor Leon Lack, from the School of Psychology at Adelaide's Flinders University, has compared 5-, 10-, 20- and 30-minute naps; he measured sleepiness, reaction time and cognitive performance before and immediately after a nap and again during the next three hours. "The 5-minute nap delivered very few benefits," says Lack. "The 20- and 30-minute naps produced improvements but the subjects took at least half an hour to wake up completely." Lack explains that the longer the sleep, the deeper it is, which can lead to a feeling called sleep inertia. "The 10-minute nap delivered immediate benefits that lasted for two to three hours, including a small but significant increase in alertness."

But how can just 10 minutes of sleep be ensured, when not in lab conditions? Most people take 5 to 10 minutes to fall asleep so they need to lie down for a total of 20 minutes to allow for 10 to 15 minutes of sleep. First, we need to change our attitudes to rest.

Australians work an average of 1811 hours each year, according to 2005 figures from the Organisation for Economic Co-operation and Development (OECD). This is the fifth highest figure from 20 nations surveyed. In addition, Dr. Kennedy stated that 9% of 20- to 30-year-olds and 16% of 30- to 50-year-olds are reporting sleeping problems. But in Australia, a culture where doing anything at all is considered better than doing nothing, lying down for a while-in the face of deadlines and urgent requests-is regarded as unacceptable by most companies. Some companies, however, are listening to experts who advise on ways to help employees take quick naps.

Both low- and high-tech napping methods are available for those who want to try. Low-tech napping can include the use of a basic relaxation room, or, in the case of the strategic public relations firm Wordplay, a 'CushoBed'. This combination of a very large cushion and a couch provides Wordplay's staff with a comfortable place to curl up for a short sleep. Then there's the high-tech 'TechnoSnooze,' an up-market sleeping pod that arrived in Australia from New York earlier this year, and which has been leased out to several companies on trial, including advertising agencies State Right Australia and Instant Publicity. Looking like a space-age reclining armchair, the TechnoSnooze has a rounded hood that lowers over the head and headphones that play relaxation music. The pod inclines forward to allow for easy entry, then reclines so that the user's feet are slightly elevated. This promotes blood circulation and reduces pressure on the lower back. After 20 minutes, the pod vibrates gently to wake you.

Harry Baker, the managing director of another large company, doesn't need such a high-tech approach. He makes good use of his media company's meditation room, which includes quiet music, candles, and incense. He encourages his staff to use it too. 'Napping is a good idea,' says Baker. 'It's like a traffic signal that slows down your brain.'

However, employees need strong workplace support from their bosses and co-workers to feel they have permission for a mini-sleep as a regular part of their working day. 'Workplace napping made a huge amount of sense to me very quickly, and I assumed the idea would sell itself, but that wasn't always the case,' says Kevin Hopkins from State Right Australia, which trialled pod-style napping for a month. 'For napping to be beneficial, you need to ensure good briefing of the managers so they are clear about the positive outcomes and are equipped to endorse, role-model, and support staff, since staff will usually take their lead from managers.' He says staff response was positive: 43% of those who booked themselves in for a pod nap said they felt 'good' and 21% 'excellent' afterwards.

*Questions 1-4*

Do the following statements agree with the information given in Reading Passage 1?

*In boxes 1-4 on your answer sheet, write*

| | |
|---|---|
| ***TRUE*** | *if the statement agrees with the information* |
| ***FALSE*** | *if the statement contradicts the information* |
| ***NOT GIVEN*** | *if there is no information on this* |

**1** The majority of mistakes in the workplace happen in the afternoon.

**2** A short nap of five minutes is enough to reduce errors at work.

**3** People who work long hours are more likely to have sleeping problems.

**4** Doing nothing is acceptable in Australian culture.

*Questions 5-10*

Complete the table below.

*Choose* ***NO MORE THAN TWO WORDS AND/OR A NUMBER*** *from the passage for each answer.*

*Write your answers in boxes 5-10 on your answer sheet.*

| Where nap takes place | Used by | Description |
|---|---|---|
| CushoBed | Wordplay | Blends features of a giant **5** ..................... and a sofa. |
| TechnoSnooze | State Right Australia<br>Instant Publicity | An ultra-modern lounger with a **6** ..................... at the top.<br>Lessens stress on the **7** ..................... |
| **9** ..................... | Harry Baker | Allows you to sleep for a maximum of **8** .....................<br>A sweet-smelling room with subtle lighting and background **10** ..................... |

*Questions 11-13*

*Answer the questions below. Choose* ***NO MORE THAN THREE WORDS*** *from the passage for each answer.*

*Write your answers in boxes 11-13 on your answer sheet.*

**11** To what does Harry Baker compare napping?

**12** Apart from their superiors, who do workers need consent from if they are to feel comfortable taking a nap at work?

**13** What do managers need to understand before they can support their staff in 'sleeping on the job'?

## READING PASSAGE 2

*You should spend about 20 minutes on **Questions 14-26,** which are based on Reading Passage 2 below.*

# Reducing the effects of jet lag

**A** We like to think we have control over our bodies, but the opposite is often true. Such is the case with 'circadian desynchrony', a condition more commonly known as jet lag. Exhaustion, headaches, difficulty concentrating, light-headedness, trouble falling asleep or staying asleep; these are common effects of jet lag. They have the power to ruin a vacation or business trip unless you learn how to trick your own body.

**B** Experimental psychologist John Caldwell has spent most of his career researching the effects of sleep deprivation and sleep restriction, while also studying countermeasures that sleep-deprived people can use to function better. Much of his research was conducted within the military community. Caldwell explains that while our bodies are able to adjust to about one time zone change per day, jet lag sets in when we cross three or more of them because it creates chaos in circadian rhythms (otherwise known as our body clock). That's a fairly new phenomenon, historically speaking. 'People now can fly from New York to Paris in nine, ten hours, whereas in 1923 you did it on a ship and it took you six days to get over to Europe,' Caldwell says. 'We just haven't evolved to the point where we can rapidly change those rhythms, because it's a relatively recent thing.'

**C** Because of the problems your body has naturally adapting to time-zone changes, you need to manually adjust your body clock, and that means changing your bedtime to be better matched with the destination to which you're traveling. Ranit Mishori, a professor of family medicine at Georgetown University School of Medicine, travels frequently to Europe, Africa, and the Middle East. To be ready to work when she arrives, she starts adjusting her bedtime two to five days in advance to match the local time at her destination. 'That means going to bed earlier when going east and waking up much earlier,' she says. When she returns to the US, she does the same but in reverse.

John Caldwell creates a timetable that includes meetings, bedtimes, and social activities so that he can easily see what time it is at home and at his destination and plan accordingly. 'A lot of times, when you look at that table, right away you're going to see where you're going to have your biggest problems,' he says. If he's just traveling for a quick business trip and will only be gone a couple of days, he avoids gradual adjustment. Instead, he tries to schedule any meetings at a time when he would be awake and focused back home.

**D** Circadian rhythms are influenced by natural light. While travel may disrupt those rhythms, you can help get them on track by regulating the amount of light that your body encounters, says Pradeep Bollu, associate director of the University of Missouri Health Care Sleep Disorders Center. When traveling east, your biological clock will be behind: '... avoiding bright light in the evening can help with advancing our biological clock,' he says. 'Similarly, bright light ... after waking up also will help advance our biological clock to suit the new time zone.' When traveling westward, he adds, the biological clock is ahead of the latest time zone. He suggests gravitating toward bright light in the evening, if possible, and exercising to stay awake later and sleep longer.

**E** One suggestion that is sometimes made is taking the hormone melatonin, which is a substance that is produced every night by the human body and helps you sleep. 'Taking a very small dose helps to recalibrate its release so that it is in sync with the time zone of your destination,' says Kem Singh, a spine surgeon in Chicago with Midwest Orthopaedics at Rush. Singh says he takes five milligrams of melatonin - which you can buy in pill form in supermarkets and many mainstreet stores in the US - on the plane and then again when he lands.

**F** Having a glass of wine on the plane may sound tempting, but it could negatively impact your sleep, which could worsen jet lag, says Quay Snyder, president and CEO of Aviation Medicine Advisory Service of Centennial, who advises pilots on staying in top condition while in the air. 'It definitely has a sedating effect as far as getting someone to sleep, but it destroys their rapid eye movement (REM) sleep so their actual mental recovery is reduced,' he says. Instead, he says, be sure and have plenty of water so that you stay hydrated while traveling.

**G** Bruce Stephen Rashbaum, owner and medical director of Capital Center for Travel and Tropical Medicine in the District of Columbia, regularly advises patients on jet lag. He considers prednisone, which is a powerful prescription medication, to be the most effective tool for jet lag recovery. He instructs patients to take the medication when they land, which is typically early in the morning, and again in the late afternoon and the next day. 'It is this simple ritual that works nearly every time,' he says. So if in doubt, you can always ask your doctor for some assistance.

Everyone responds to jet lag differently. For those who suffer, the first week will be the most challenging, but after that, your body should start to recover.

*Questions 14-20*

Reading Passage 2 has six paragraphs, **A-G.**

*Choose the correct heading for each paragraph from the list of headings below.*

*Write the correct number, i-ix, in boxes 14-20 on your answer sheet.*

| **List of Headings** | |
|---|---|
| **i** | Requesting help from professionals |
| **ii** | Types of meals and beverages that help |
| **iii** | What not to do on a flight |
| **iv** | Symptoms of jet lag |
| **v** | Altering your sleep schedules |
| **vi** | Types of exercise to do |
| **vii** | A problem of the modern age |
| **viii** | A remedy available from ordinary shops |
| **ix** | Timing exposure to sunshine |

**14** Paragraph **A**

**15** Paragraph **B**

**16** Paragraph **C**

**17** Paragraph **D**

**18** Paragraph **E**

**19** Paragraph **F**

**20** Paragraph **G**

*Questions 21-23*

*Look at the following statements (Questions 21-25) and the list of experts below.*

*Match each statement with the correct expert,* ***A-F.***

*Write the correct letter,* ***A-F,*** *in boxes* ***21-23*** *on your answer sheet.*

**21** Using strong medicine is the most efficient way to get over jet lag.

**22** Using natural supplements to reset biological processes can help travelers.

**23** Having certain types of drinks lessens the quality of sleep.

**List of Experts**

| | |
|---|---|
| **A** | John Caldwell |
| **B** | Ranit Mishori |
| **C** | Pradeep Bollu |
| **D** | Kern Singh |
| **E** | Quay Snyder |
| **F** | Bruce Stephen Rashbaum |

*Questions 24-26*

*Complete the summary below.*

*Choose **ONE WORD ONLY** from the passage for each answer.*

*Write your answers in boxes 24-26 on your answer sheet.*

## Why we experience jet lag

John Caldwell has studied sleep issues among **24** ..................... personnel. He explains that jet lag is an issue because the human body can only naturally adapt to one change in time zones per day; traveling over more than that causes **25** ..................... for our bodies. Unlike when traveling by **26** ....................., flying has resulted in us being able to change many time zones quickly. As this is a fairly new form of travel, the human body hasn't yet evolved to be able to cope with this.

## READING PASSAGE 3

*You should spend about 20 minutes on **Questions 27-40,** which are based on Reading Passage 3 below.*

# Research into taming a fox

*How to Tame a Fox (and Build a Dog)* is written by Lee Alan Dugatkin, an American evolutionary biologist, and Lyudmila Trut, a Russian professor of evolutionary genetics. The book describes the scientific effort to domesticate Siberian silver foxes - a project that Trut has been involved in as lead researcher for more than 50 years. When the project began, its purpose was to shed light on how dogs had gradually evolved from wolves, and silver foxes were chosen because they are the wolf's close genetic cousins.

There is much material to cover in the book, but the authors condense this without sacrificing any significant details of the project. The introduction reveals the end product of the experiment by describing the behavior of some very tame foxes that willingly interact with humans. This revelation, however, does nothing to lessen the reader's fascination. Indeed, the authors do an excellent job of expressing the joy of discovery of those early years of genetic research.

Interwoven with descriptions of the project and the astonishing changes in the foxes are the personal stories of the people involved, an approach rarely taken in scientific books, but one which works extremely well here. A major figure to appear in the book is geneticist Dmitri Belyaev, who first set up the domestication experiment in 1952, and who invited Trut to join the project in 1959. Trut continues his work today at their research facility in Siberia, where winter temperatures reach -40°C. Her team endures the harsh conditions because they obviously care deeply for the science and the animals they are in charge of

The evolution of the wolf into a dog has long puzzled scientists. How did wolves first come in contact with humans? Why were some wolves less averse to humans than others? To answer these questions, Belyaev devised an experiment to 'breed the wild out of the animal' on an accelerated timeline.

As lead scientist at the Central Research Laboratory on Fur Breeding Animals in Moscow, Belyaev was less interested in the official line of research and more interested in the impact of domestication on animal genomes. After moving to Siberia, he obtained some foxes from a fox farm for a breeding experiment. He chose less aggressive foxes that did not immediately try to bite their caretakers. At the earliest stage of the experiment, he felt it was necessary to hide his activities from his superiors at the Central Research Laboratory.

Initially, 12 foxes were chosen, and within three breeding seasons the pilot study produced preliminary results. Some pups were not consistently aggressive; sometimes they even appeared indifferent toward people. These results were enough for Belyaev to hire Trut, then studying animal behavior at Moscow State University. Her job was to select the foxes for breeding and record details of changes. Among the eighth-generation foxes, a couple were calm enough for Trut to pick up. These, along with others that were slightly calmer than the rest, were selected for breeding as part of a larger project.

Repeating the selection for the tamest foxes in each succeeding generation, Trut and Belyaev noticed new behaviors. The first milestone occurred with a fox named Ember. Tail Wagging had never been recorded in foxes in the wild or captivity; it is the behaviour of dogs signalling happiness. One day Ember wagged his tail when Trut approached. He was the only pup of his generation to do so, but this generation was also markedly calmer than any previous one.

By the sixth generation of the larger project, pups displayed new behaviours. They pressed against the front of their crates seeking human contact, licked hands, and whined when people left. Fox pups whine for food from their mothers but had never done so for humans. Nor had they been documented licking hands. The pups exhibiting these behaviours were put in an elite group for breeding. Pups that did not exhibit these behaviours, or only occasionally exhibited a few, were maintained as a control group.

Belyaev and Trut made further observations. Instead of standing straight up, the elite pups' ears flopped over. Their snouts shortened, and some developed curly tails. These traits made them more visually appealing to humans. In short, over the generations, the foxes began to resemble dogs. It was then discovered that domestication creates complex interactions between genes that influence physical traits and genes that influence behaviour. But how did this domestication happen with ancient wolves? Belyaev theorised that naturally calmer wolves approached people in search of pieces of animal meat and bone they had discarded. As these calmer wolves interbred, the animals were no longer aggressive towards humans. This theory makes more sense than older ideas, which proposed that neolithic people stole wolf cubs so they could be trained to assist with practical tasks such as shepherding, guarding and hunting. For Belyaev and Trut, their new theory answered the question of how we got from wolf to dog, but not how we then got to a faithful four-legged family member.

Trut proposed to further chart the domestication of foxes by raising a fox called Pushinka at home. In time, Pushinka began differentiating between Trut and her colleagues, and other people who only seldom called at the house. Pushinka and her pups announced strangers' arrival with a bark no other fox had exhibited. So far 57 generations of foxes have been bred, and researchers have begun recording and decoding the vocalisations they have developed.

*Questions 27-31*

Do the following statements agree with the claims of the writer in Reading Passage 3?

*In boxes 27-31 on your answer sheet, write*

| | |
|---|---|
| ***YES*** | *if the statement agrees with the claims of the writer* |
| ***NO*** | *if the statement contradicts the claims of the writer* |
| ***NOT GIVEN*** | *if it is impossible to say what the writer thinks about this* |

**27** Trut could never have imagined that the project would continue for over 50 years.

**28** The authors have successfully conveyed all the key facts and important data associated with the research.

**29** It is a shame that the outcome of the experiment is revealed in the book's introduction.

**30** The authors made a mistake by including personal stories in the book.

**31** It is clear that the staff in the Siberian research facility have genuine concern for their foxes.

*Questions 32-34*

*Choose the correct letter, **A, B,** **C,** or **D.***

*Write the correct letter in boxes 32-34 on your answer sheet.*

**32** What is the writer doing in the fourth paragraph?

**A** examining Belyaev's approach to proving a theory
**B** comparing the behavioural traits of wolves to foxes giving
**C** the background to Belyaev's research project challenging
**D** traditional views on how species evolved

**33** After being assigned a project by the Central Research Laboratory, Belyaev responded by

**A** resigning and applying for work in another institution.
**B** keeping secret the work he was actually carrying out.
**C** choosing subjects that were hostile to people. seeking
**D** alternative means of funding for his scheme.

**34** What are we told about Lyudmila Trut in the sixth paragraph?

**A** Two of Belyaev's foxes seemed willing to let her handle them.
**B** She disagreed with Belyaev about the future direction of the project. She
**C** had worked with foxes before being employed by Belyaev. Belyaev had
**D** been an influence on her during her degree course.

*Questions 35-40*

*Complete the summary using the list of words, **A-I,** below.*

*Write the correct letter, **A-I,** in boxes 35-40 on your answer sheet.*

## Belyaev and Trut's observations and theories

Belyaev and Trut observed that the elite group of foxes started to have a range of **35** ..................... in common with dogs. Believing that domestication was responsible for this, Belyaev then proposed a theory about domestication in ancient wolves. Whereas other researchers had suggested early man had taken wolf cubs to turn them into **36** ....................., Belyaev thought the wolves had associated human groups with **37** ...................... This would explain the initial contact that wolves had with humans, but not how dogs eventually developed as **38** ......................

To see how this might have occurred, Trut raised a fox called Pushinka and her pups in her own home. In time, when **39** ..................... came to Trut's house, Pushinka and the pups could tell the difference between them and those they were more familiar with. Currently, Trut and her team are analysing the **40** ..................... produced by the latest generation of foxes.

| | | | | | |
|---|---|---|---|---|---|
| **A** | useful workers | **B** | loyal pets physical | **C** | colour patterns |
| **D** | specific breeds | **E** | features warning | **F** | new sounds |
| **G** | irregular visitors | **H** | signs | **I** | food sources |